

SHAKTICOP

ENTERPRISE WOMEN SAFETY PLATFORM

Software Requirements Specification (SRS)

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Chapter 1: Executive Summary

This Software Requirements Specification (SRS) establishes the requirements and functional design for the ShaktiCop platform. Prepared independently by Adarsh Singh, this document is a comprehensive software design and requirements proposal designed for submission and review by Uttar Pradesh Police. The platform bridges critical safety response infrastructure with vocational empowerment programs (Mission Shakti), enabling integrated, audited, and GIGW 3.0 compliant security workflows.

The platform represents a shift in how women's safety is administered at the grassroots level. By merging rapid distress response mechanisms directly with local community development and vocational tools, ShaktiCop addresses both the immediate emergency need and the long-term empowerment objectives mandated by the State's flagship Mission Shakti initiative. This documentation is structured to guide the engineering phase directly into development and deployment.

Chapter 2: Introduction

Section 2.1: Purpose

The purpose of this specification is to define the functional, performance, security, and interface requirements for the ShaktiCop platform. It is designed to act as the primary blueprint for implementation partners.

Section 2.2: Scope

ShaktiCop encompasses both safety (distress triggers, PCR dispatch queues, silent SMS telemetry routing) and empowerment modules (counseling sessions, shelter inventory trackers, self-defense classes, and JSS vocational training). Integration endpoints, CCTNS backend sync routes, National ERSS (112 India) configurations, and DigiLocker APIs are to be finalized in consultation with Uttar Pradesh Police during implementation.

Section 2.3: Definitions, Acronyms and Abbreviations

- SOS: Save Our Souls (Emergency Distress Signal)
- PCR: Patrol Car Response Unit
- CCTNS: Crime and Criminal Tracking Network & Systems
- ERSS: Emergency Response Support System (112 India)
- JSS: Jan Shikshan Sansthan (Vocational Skill Development Initiative)
- DPDP: Digital Personal Data Protection Act, 2023
- DBT: Direct Benefit Transfer (Aadhaar-linked welfare payouts)
- SHO: Station House Officer (Officer-in-charge of a local Police Station)
- SLA: Service Level Agreement (Target response times for dispatch units)

Chapter 3: Business Problem & Gap Analysis

Current safety response systems operate in silos. Emergency calls (112) are detached from domestic violence shelter tracking, making it difficult to coordinate help. When a woman triggers distress in a remote village, the dispatch center lacks access to nearby shelter availability or assigned female officer contacts. Furthermore, legal aid registration and confidential counseling schedules are paper-based, leading to tracking errors and privacy exposures.

Additionally, vocational rehabilitation and self-defense training registries are managed independently. This separation prevents police command consoles from tracking the complete rehabilitation cycle of a victim. ShaktiCop resolves these gaps by integrating all safety registries, dispatch telemetry, counseling portals, and skill courses into a single secure platform.

Chapter 4: Proposed Solution & Core Pillars

ShaktiCop delivers a unified platform that acts as the single source of truth for women's safety. It consists of four core pillars:

1. Immediate Emergency Distress: Integrates a silent SOS tracking system, automatic location updates, and PCR vehicle matching.
2. Confidential Triage and Complaint Filing: Supports voice-recorded audio complaints that transcribe local dialects.
3. Empowerment and Vocational Support: Connects citizens with JSS skill courses and self-defense workshops.
4. Unified Command Dashboard: Provides police and administrators with live hotspot heatmaps and response SLA statistics.

Chapter 5: Governance & User Roles

The system defines authorization boundaries for 8 key user roles:

1. Citizen (Women): Triggers SOS, records complaints, tracks travel path.
2. Police Officer (PCR Driver): Navigates to incident location, updates status.
3. Police Station Admin (SHO): Audits local complaints, manages Anti-Romeo patrols.
4. District Admin (SP): Reviews response SLAs, hotspot heatmaps.
5. State Admin (DGP): Accesses aggregate safety statistics.
6. NGO Partner: Manages shelter beds, monitors rehabilitation progress.
7. Counsellor: Holds confidential therapeutic calls.
8. Super Admin: System configurations, database archival, audits.

Chapter 6: Functional Requirements by Module

Section 6.1: Registration

Enables citizens to register online using mobile authentication and link profiles securely.

- Purpose: Facilitate secure operational transactions for this module.
- Business Objective: Enforce safety regulations and reduce dispatch response SLAs.
- Actor Groups: Citizen, Police Officer, Station Admin, NGO, Counsellor, SP, DGP.
- Preconditions: Active session, mobile validation complete, consent logged.
- Postconditions: State machine updated, notifications sent, auditable ledger written.
- Main Flow: Action triggered -> Validation passed -> API query executed -> Database committed -> Response rendered.
- Backend Logic: State checks verify operator scopes before writing changes.
- Database Impact: Writes transaction logs to audit ledgers.
- Integration Details: To be finalized with Uttar Pradesh Police during implementation.

Section 6.2: Login

Enables role-based access for citizens, police officers, administrators, and coordinators.

- Purpose: Facilitate secure operational transactions for this module.
- Business Objective: Enforce safety regulations and reduce dispatch response SLAs.
- Actor Groups: Citizen, Police Officer, Station Admin, NGO, Counsellor, SP, DGP.
- Preconditions: Active session, mobile validation complete, consent logged.
- Postconditions: State machine updated, notifications sent, auditable ledger written.
- Main Flow: Action triggered -> Validation passed -> API query executed -> Database committed -> Response rendered.
- Backend Logic: State checks verify operator scopes before writing changes.
- Database Impact: Writes transaction logs to audit ledgers.
- Integration Details: To be finalized with Uttar Pradesh Police during implementation.

Section 6.3: OTP Verification

Verifies authentication sessions via dynamic Aadhaar-linked OTP codes.

- Purpose: Facilitate secure operational transactions for this module.
- Business Objective: Enforce safety regulations and reduce dispatch response SLAs.
- Actor Groups: Citizen, Police Officer, Station Admin, NGO, Counsellor, SP, DGP.
- Preconditions: Active session, mobile validation complete, consent logged.
- Postconditions: State machine updated, notifications sent, auditable ledger written.
- Main Flow: Action triggered -> Validation passed -> API query executed -> Database committed -> Response rendered.
- Backend Logic: State checks verify operator scopes before writing changes.
- Database Impact: Writes transaction logs to audit ledgers.
- Integration Details: To be finalized with Uttar Pradesh Police during implementation.

Section 6.4: User Profile

Manages user-specific preferences, trusted contacts, and data consent ledgers.

- Purpose: Facilitate secure operational transactions for this module.
- Business Objective: Enforce safety regulations and reduce dispatch response SLAs.

- Actor Groups: Citizen, Police Officer, Station Admin, NGO, Counsellor, SP, DGP.
- Preconditions: Active session, mobile validation complete, consent logged.
- Postconditions: State machine updated, notifications sent, auditable ledger written.
- Main Flow: Action triggered -> Validation passed -> API query executed -> Database committed -> Response rendered.
- Backend Logic: State checks verify operator scopes before writing changes.
- Database Impact: Writes transaction logs to audit ledgers.
- Integration Details: To be finalized with Uttar Pradesh Police during implementation.

Section 6.5: SOS Distress Tracker

Covert SOS activation (triple power button taps). Transmits location coordinate updates to the dispatch console every 10 seconds.

- Purpose: Facilitate secure operational transactions for this module.
- Business Objective: Enforce safety regulations and reduce dispatch response SLAs.
- Actor Groups: Citizen, Police Officer, Station Admin, NGO, Counsellor, SP, DGP.
- Preconditions: Active session, mobile validation complete, consent logged.
- Postconditions: State machine updated, notifications sent, auditable ledger written.
- Main Flow: Action triggered -> Validation passed -> API query executed -> Database committed -> Response rendered.
- Backend Logic: State checks verify operator scopes before writing changes.
- Database Impact: Writes transaction logs to audit ledgers.
- Integration Details: To be finalized with Uttar Pradesh Police during implementation.

Section 6.6: Emergency Complaint

Logs domestic abuse complaints and alerts nearby station units.

- Purpose: Facilitate secure operational transactions for this module.
- Business Objective: Enforce safety regulations and reduce dispatch response SLAs.
- Actor Groups: Citizen, Police Officer, Station Admin, NGO, Counsellor, SP, DGP.
- Preconditions: Active session, mobile validation complete, consent logged.
- Postconditions: State machine updated, notifications sent, auditable ledger written.
- Main Flow: Action triggered -> Validation passed -> API query executed -> Database committed -> Response rendered.
- Backend Logic: State checks verify operator scopes before writing changes.
- Database Impact: Writes transaction logs to audit ledgers.
- Integration Details: To be finalized with Uttar Pradesh Police during implementation.

Section 6.7: Voice Complaint & Transcription

Enables voice-recorded audio complaints. Integrates a local dialect translator to assist in case registration.

- Purpose: Facilitate secure operational transactions for this module.
- Business Objective: Enforce safety regulations and reduce dispatch response SLAs.

- Actor Groups: Citizen, Police Officer, Station Admin, NGO, Counsellor, SP, DGP.
- Preconditions: Active session, mobile validation complete, consent logged.
- Postconditions: State machine updated, notifications sent, auditable ledger written.
- Main Flow: Action triggered -> Validation passed -> API query executed -> Database committed -> Response rendered.
- Backend Logic: State checks verify operator scopes before writing changes.
- Database Impact: Writes transaction logs to audit ledgers.
- Integration Details: To be finalized with Uttar Pradesh Police during implementation.

Section 6.8: Evidence Upload

Supports uploading photos and videos linked to active complaint cases, enforcing SHA-256 integrity validation.

- Purpose: Facilitate secure operational transactions for this module.
- Business Objective: Enforce safety regulations and reduce dispatch response SLAs.
- Actor Groups: Citizen, Police Officer, Station Admin, NGO, Counsellor, SP, DGP.
- Preconditions: Active session, mobile validation complete, consent logged.
- Postconditions: State machine updated, notifications sent, auditable ledger written.
- Main Flow: Action triggered -> Validation passed -> API query executed -> Database committed -> Response rendered.
- Backend Logic: State checks verify operator scopes before writing changes.
- Database Impact: Writes transaction logs to audit ledgers.
- Integration Details: To be finalized with Uttar Pradesh Police during implementation.

Section 6.9: Safe Travel

Tracks public transit rides (autos, cabs, trains). Alerts trusted contacts and the local patrol squad on route deviations.

- Purpose: Facilitate secure operational transactions for this module.
- Business Objective: Enforce safety regulations and reduce dispatch response SLAs.
- Actor Groups: Citizen, Police Officer, Station Admin, NGO, Counsellor, SP, DGP.
- Preconditions: Active session, mobile validation complete, consent logged.
- Postconditions: State machine updated, notifications sent, auditable ledger written.
- Main Flow: Action triggered -> Validation passed -> API query executed -> Database committed -> Response rendered.
- Backend Logic: State checks verify operator scopes before writing changes.
- Database Impact: Writes transaction logs to audit ledgers.
- Integration Details: To be finalized with Uttar Pradesh Police during implementation.

Section 6.10: Live Tracking

Visualizes real-time coordinate updates on central command maps.

- Purpose: Facilitate secure operational transactions for this module.
- Business Objective: Enforce safety regulations and reduce dispatch response SLAs.

- Actor Groups: Citizen, Police Officer, Station Admin, NGO, Counsellor, SP, DGP.
- Preconditions: Active session, mobile validation complete, consent logged.
- Postconditions: State machine updated, notifications sent, auditable ledger written.
- Main Flow: Action triggered -> Validation passed -> API query executed -> Database committed -> Response rendered.
- Backend Logic: State checks verify operator scopes before writing changes.
- Database Impact: Writes transaction logs to audit ledgers.
- Integration Details: To be finalized with Uttar Pradesh Police during implementation.

Section 6.11: Emergency Contacts

Allows citizens to define and authorize contact details for automated SOS alerts.

- Purpose: Facilitate secure operational transactions for this module.
- Business Objective: Enforce safety regulations and reduce dispatch response SLAs.
- Actor Groups: Citizen, Police Officer, Station Admin, NGO, Counsellor, SP, DGP.
- Preconditions: Active session, mobile validation complete, consent logged.
- Postconditions: State machine updated, notifications sent, auditable ledger written.
- Main Flow: Action triggered -> Validation passed -> API query executed -> Database committed -> Response rendered.
- Backend Logic: State checks verify operator scopes before writing changes.
- Database Impact: Writes transaction logs to audit ledgers.
- Integration Details: To be finalized with Uttar Pradesh Police during implementation.

Section 6.12: Nearby Police Station

Uses device GPS to locate and map nearby station beats.

- Purpose: Facilitate secure operational transactions for this module.
- Business Objective: Enforce safety regulations and reduce dispatch response SLAs.
- Actor Groups: Citizen, Police Officer, Station Admin, NGO, Counsellor, SP, DGP.
- Preconditions: Active session, mobile validation complete, consent logged.
- Postconditions: State machine updated, notifications sent, auditable ledger written.
- Main Flow: Action triggered -> Validation passed -> API query executed -> Database committed -> Response rendered.
- Backend Logic: State checks verify operator scopes before writing changes.
- Database Impact: Writes transaction logs to audit ledgers.
- Integration Details: To be finalized with Uttar Pradesh Police during implementation.

Section 6.13: Shelter Homes

Registers safe shelters and synchronizes bed capacity counts.

- Purpose: Facilitate secure operational transactions for this module.
- Business Objective: Enforce safety regulations and reduce dispatch response SLAs.
- Actor Groups: Citizen, Police Officer, Station Admin, NGO, Counsellor, SP, DGP.
- Preconditions: Active session, mobile validation complete, consent logged.

- Postconditions: State machine updated, notifications sent, auditable ledger written.
- Main Flow: Action triggered -> Validation passed -> API query executed -> Database committed -> Response rendered.
- Backend Logic: State checks verify operator scopes before writing changes.
- Database Impact: Writes transaction logs to audit ledgers.
- Integration Details: To be finalized with Uttar Pradesh Police during implementation.

Section 6.14: Counselling Portal

Facilitates booking private sessions with trauma and domestic abuse counselors.

- Purpose: Facilitate secure operational transactions for this module.
- Business Objective: Enforce safety regulations and reduce dispatch response SLAs.
- Actor Groups: Citizen, Police Officer, Station Admin, NGO, Counsellor, SP, DGP.
- Preconditions: Active session, mobile validation complete, consent logged.
- Postconditions: State machine updated, notifications sent, auditable ledger written.
- Main Flow: Action triggered -> Validation passed -> API query executed -> Database committed -> Response rendered.
- Backend Logic: State checks verify operator scopes before writing changes.
- Database Impact: Writes transaction logs to audit ledgers.
- Integration Details: To be finalized with Uttar Pradesh Police during implementation.

Section 6.15: Skill Development (JSS)

Enables enrollment in vocational courses (Digital Literacy, Beauty & Wellness). Connects with Etawah branches.

- Purpose: Facilitate secure operational transactions for this module.
- Business Objective: Enforce safety regulations and reduce dispatch response SLAs.
- Actor Groups: Citizen, Police Officer, Station Admin, NGO, Counsellor, SP, DGP.
- Preconditions: Active session, mobile validation complete, consent logged.
- Postconditions: State machine updated, notifications sent, auditable ledger written.
- Main Flow: Action triggered -> Validation passed -> API query executed -> Database committed -> Response rendered.
- Backend Logic: State checks verify operator scopes before writing changes.
- Database Impact: Writes transaction logs to audit ledgers.
- Integration Details: To be finalized with Uttar Pradesh Police during implementation.

Section 6.16: Community Forum

Allows moderated group discussions on rights and safety topics.

- Purpose: Facilitate secure operational transactions for this module.
- Business Objective: Enforce safety regulations and reduce dispatch response SLAs.
- Actor Groups: Citizen, Police Officer, Station Admin, NGO, Counsellor, SP, DGP.
- Preconditions: Active session, mobile validation complete, consent logged.
- Postconditions: State machine updated, notifications sent, auditable ledger written.

- Main Flow: Action triggered -> Validation passed -> API query executed -> Database committed -> Response rendered.
- Backend Logic: State checks verify operator scopes before writing changes.
- Database Impact: Writes transaction logs to audit ledgers.
- Integration Details: To be finalized with Uttar Pradesh Police during implementation.

Section 6.17: Chatbot (ShaktiCop AI)

A virtual assistant resolving helpline queries and booking sessions.

- Purpose: Facilitate secure operational transactions for this module.
- Business Objective: Enforce safety regulations and reduce dispatch response SLAs.
- Actor Groups: Citizen, Police Officer, Station Admin, NGO, Counsellor, SP, DGP.
- Preconditions: Active session, mobile validation complete, consent logged.
- Postconditions: State machine updated, notifications sent, auditable ledger written.
- Main Flow: Action triggered -> Validation passed -> API query executed -> Database committed -> Response rendered.
- Backend Logic: State checks verify operator scopes before writing changes.
- Database Impact: Writes transaction logs to audit ledgers.
- Integration Details: To be finalized with Uttar Pradesh Police during implementation.

Section 6.18: AI Triage Assistant

Prioritizes and routes emergency alerts to dispatcher queues based on keyword severity.

- Purpose: Facilitate secure operational transactions for this module.
- Business Objective: Enforce safety regulations and reduce dispatch response SLAs.
- Actor Groups: Citizen, Police Officer, Station Admin, NGO, Counsellor, SP, DGP.
- Preconditions: Active session, mobile validation complete, consent logged.
- Postconditions: State machine updated, notifications sent, auditable ledger written.
- Main Flow: Action triggered -> Validation passed -> API query executed -> Database committed -> Response rendered.
- Backend Logic: State checks verify operator scopes before writing changes.
- Database Impact: Writes transaction logs to audit ledgers.
- Integration Details: To be finalized with Uttar Pradesh Police during implementation.

Section 6.19: Notification Services

Routes push alerts, SMS notifications, and WhatsApp gateway dispatches.

- Purpose: Facilitate secure operational transactions for this module.
- Business Objective: Enforce safety regulations and reduce dispatch response SLAs.
- Actor Groups: Citizen, Police Officer, Station Admin, NGO, Counsellor, SP, DGP.
- Preconditions: Active session, mobile validation complete, consent logged.
- Postconditions: State machine updated, notifications sent, auditable ledger written.
- Main Flow: Action triggered -> Validation passed -> API query executed -> Database committed -> Response rendered.

- Backend Logic: State checks verify operator scopes before writing changes.
- Database Impact: Writes transaction logs to audit ledgers.
- Integration Details: To be finalized with Uttar Pradesh Police during implementation.

Section 6.20: Reports Generator

Compiles daily and monthly ticket metrics by district.

- Purpose: Facilitate secure operational transactions for this module.
- Business Objective: Enforce safety regulations and reduce dispatch response SLAs.
- Actor Groups: Citizen, Police Officer, Station Admin, NGO, Counsellor, SP, DGP.
- Preconditions: Active session, mobile validation complete, consent logged.
- Postconditions: State machine updated, notifications sent, auditable ledger written.
- Main Flow: Action triggered -> Validation passed -> API query executed -> Database committed -> Response rendered.
- Backend Logic: State checks verify operator scopes before writing changes.
- Database Impact: Writes transaction logs to audit ledgers.
- Integration Details: To be finalized with Uttar Pradesh Police during implementation.

Section 6.21: Analytics Heatmaps

Aggregates incident markers to map district safety ratings and vulnerabilities.

- Purpose: Facilitate secure operational transactions for this module.
- Business Objective: Enforce safety regulations and reduce dispatch response SLAs.
- Actor Groups: Citizen, Police Officer, Station Admin, NGO, Counsellor, SP, DGP.
- Preconditions: Active session, mobile validation complete, consent logged.
- Postconditions: State machine updated, notifications sent, auditable ledger written.
- Main Flow: Action triggered -> Validation passed -> API query executed -> Database committed -> Response rendered.
- Backend Logic: State checks verify operator scopes before writing changes.
- Database Impact: Writes transaction logs to audit ledgers.
- Integration Details: To be finalized with Uttar Pradesh Police during implementation.

Section 6.22: Settings Manager

Enables operators to modify system parameters and key rotation cycles.

- Purpose: Facilitate secure operational transactions for this module.
- Business Objective: Enforce safety regulations and reduce dispatch response SLAs.
- Actor Groups: Citizen, Police Officer, Station Admin, NGO, Counsellor, SP, DGP.
- Preconditions: Active session, mobile validation complete, consent logged.
- Postconditions: State machine updated, notifications sent, auditable ledger written.
- Main Flow: Action triggered -> Validation passed -> API query executed -> Database committed -> Response rendered.
- Backend Logic: State checks verify operator scopes before writing changes.

- Database Impact: Writes transaction logs to audit ledgers.
- Integration Details: To be finalized with Uttar Pradesh Police during implementation.

Chapter 7: Interface & Screen Specifications

The following specifications document the exact layout, fields, validations, and transitions identified within the ShaktiCop index.html prototype.

Section 7.1: Screen Layout for Citizen Home Screen

Main navigation panel containing service category tabs (Mission Shakti and Police Services) and a float chatbot FAB.

- Inputs: Standard forms, selections, text areas.
- Buttons & Controls: Glassmorphism styles with hover scaling effects.
- Client Validations: Checks fields for blanks, non-digit mobile entries, and invalid selectors.
- API Queries: To be finalized with Uttar Pradesh Police during implementation.

Section 7.2: Screen Layout for OTP Login Panel

Modal overlay validating a 10-digit mobile number and verified 4-digit OTP code (Demo bypass: 1234).

- Inputs: Standard forms, selections, text areas.
- Buttons & Controls: Glassmorphism styles with hover scaling effects.
- Client Validations: Checks fields for blanks, non-digit mobile entries, and invalid selectors.
- API Queries: To be finalized with Uttar Pradesh Police during implementation.

Section 7.3: Screen Layout for Mahila Help Desk Form

Intake form requiring District selection (Lucknow, Etawah, Kanpur) and dynamic station lists (Etawah Civil Lines).

- Inputs: Standard forms, selections, text areas.
- Buttons & Controls: Glassmorphism styles with hover scaling effects.
- Client Validations: Checks fields for blanks, non-digit mobile entries, and invalid selectors.
- API Queries: To be finalized with Uttar Pradesh Police during implementation.

Section 7.4: Screen Layout for Domestic Violence SOS Panel

Emergency modal with Red SOS trigger button and type-of-help selector (Shelter, Legal, Medical).

- Inputs: Standard forms, selections, text areas.
- Buttons & Controls: Glassmorphism styles with hover scaling effects.
- Client Validations: Checks fields for blanks, non-digit mobile entries, and invalid selectors.
- API Queries: To be finalized with Uttar Pradesh Police during implementation.

Section 7.5: Screen Layout for Safe Travel Monitor

Interactive tracker displaying journey start/end points, transport mode, and a simulated road progress bar.

- Inputs: Standard forms, selections, text areas.
- Buttons & Controls: Glassmorphism styles with hover scaling effects.
- Client Validations: Checks fields for blanks, non-digit mobile entries, and invalid selectors.
- API Queries: To be finalized with Uttar Pradesh Police during implementation.

Section 7.6: Screen Layout for Voice Complaint Recorder

Audio modal containing mic control, duration counter, dynamic CSS waveform anim, and Play preview.

- Inputs: Standard forms, selections, text areas.
- Buttons & Controls: Glassmorphism styles with hover scaling effects.
- Client Validations: Checks fields for blanks, non-digit mobile entries, and invalid selectors.
- API Queries: To be finalized with Uttar Pradesh Police during implementation.

Section 7.7: Screen Layout for Shelter Homes Directory

Filtered list rendering Nari Niketan and Sakhi OSC cards with address, phone links, and bed capacity.

- Inputs: Standard forms, selections, text areas.
- Buttons & Controls: Glassmorphism styles with hover scaling effects.
- Client Validations: Checks fields for blanks, non-digit mobile entries, and invalid selectors.
- API Queries: To be finalized with Uttar Pradesh Police during implementation.

Section 7.8: Screen Layout for Live Ride Tracker

Map-styled tracking window showing cab license plates, driver details, ETA timer, and trip share options.

- Inputs: Standard forms, selections, text areas.
- Buttons & Controls: Glassmorphism styles with hover scaling effects.
- Client Validations: Checks fields for blanks, non-digit mobile entries, and invalid selectors.
- API Queries: To be finalized with Uttar Pradesh Police during implementation.

Chapter 8: System Architecture

ShaktiCop utilizes a three-tier architecture:

1. Client Tier: Responsive HTML5/JS frontend complying with WCAG 2.1 AA.
2. Application Tier: Microservices containing SOS Triage, Chatbot, and JSS Course managers.
3. Database Tier: High-availability relational database cluster with read replicas.

Network and hardware configurations are to be finalized with Uttar Pradesh Police during implementation.

Chapter 9: Database Schema Specifications

The database layout consists of entity tables. Daily partitioning is enforced on telemetry logs. Audit entries are written to write-once-read-many (WORM) storage. All columns, datatypes, and constraints are detailed below.

Section 9.1: Schema for tbl_citizens

Column Name	Data Type	Constraints / Description
id	UUID	PRIMARY KEY, NOT NULL
mobile_number	VARCHAR(15)	UNIQUE, INDEXED, NOT NULL
aadhaar_vault_token	VARCHAR(256)	DigiLocker token. To be finalized with UP Police during implementation.
full_name	VARCHAR(100)	Encrypted, Masked
created_at	TIMESTAMP	DEFAULT NOW()
status	VARCHAR(20)	DEFAULT 'ACTIVE'

Table 1: Columns Layout for tbl_citizens

Section 9.2: Schema for tbl_police_stations

Column Name	Data Type	Constraints / Description
id	UUID	PRIMARY KEY, NOT NULL

station_name	VARCHAR(100)	INDEXED, NOT NULL
district	VARCHAR(50)	INDEXED, NOT NULL
station_code	VARCHAR(20)	UNIQUE, NOT NULL
contact_number	VARCHAR(15)	NOT NULL

Table 2: Columns Layout for tbl_police_stations

Section 9.3: Schema for tbl_police_officers

Column Name	Data Type	Constraints / Description
id	UUID	PRIMARY KEY, NOT NULL
officer_name	VARCHAR(100)	NOT NULL
badge_number	VARCHAR(30)	UNIQUE, INDEXED, NOT NULL
station_id	UUID	FOREIGN KEY references tbl_police_stations(id)
mobile_number	VARCHAR(15)	NOT NULL
assigned_role	VARCHAR(30)	e.g., Anti-Romeo, Help Desk, Patrol

Table 3: Columns Layout for tbl_police_officers

Section 9.4: Schema for tbl_complaints

Column Name	Data Type	Constraints / Description
id	UUID	PRIMARY KEY, NOT NULL
complaint_id_display	VARCHAR(15)	UNIQUE, INDEXED, NOT NULL

citizen_id	UUID	FOREIGN KEY references tbl_citizens(id)
station_id	UUID	FOREIGN KEY references tbl_police_stations(id)
audio_file_hash	VARCHAR(64)	SHA-256 for BNSS 2023 compliance
transcript_text	TEXT	Encrypted narrative text
status	VARCHAR(20)	DEFAULT 'SUBMITTED'

Table 4: Columns Layout for tbl_complaints

Section 9.5: Schema for tbl_sos_alerts

Column Name	Data Type	Constraints / Description
id	UUID	PRIMARY KEY, NOT NULL
citizen_id	UUID	FOREIGN KEY references tbl_citizens(id)
start_latitude	DECIMAL(9,6)	NOT NULL
start_longitude	DECIMAL(9,6)	NOT NULL
pcr_assigned_id	UUID	FOREIGN KEY references tbl_police_officers(id)
dispatch_status	VARCHAR(30)	DEFAULT 'TRIGGERED'

Table 5: Columns Layout for tbl_sos_alerts

Section 9.6: Schema for tbl_emergency_contacts

Column Name	Data Type	Constraints / Description
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id	UUID	PRIMARY KEY, NOT NULL
citizen_id	UUID	FOREIGN KEY references tbl_citizens(id)
contact_name	VARCHAR(100)	NOT NULL
contact_phone	VARCHAR(15)	NOT NULL
relation	VARCHAR(30)	NOT NULL

Table 6: Columns Layout for tbl_emergency_contacts

Section 9.7: Schema for tbl_vehicles

Column Name	Data Type	Constraints / Description
id	UUID	PRIMARY KEY, NOT NULL
license_plate	VARCHAR(20)	UNIQUE, INDEXED, NOT NULL
vehicle_type	VARCHAR(30)	e.g., Patrol Car, Anti-Romeo Patrol
current_lat	DECIMAL(9,6)	NULL
current_lng	DECIMAL(9,6)	NULL
active_status	VARCHAR(20)	DEFAULT 'AVAILABLE'

Table 7: Columns Layout for tbl_vehicles

Section 9.8: Schema for tbl_ngos

Column Name	Data Type	Constraints / Description
id	UUID	PRIMARY KEY, NOT NULL

ngo_name	VARCHAR(100)	NOT NULL
district	VARCHAR(50)	INDEXED, NOT NULL
contact_number	VARCHAR(15)	NOT NULL
address	TEXT	NOT NULL

Table 8: Columns Layout for tbl_ngos

Section 9.9: Schema for tbl_counsellors

Column Name	Data Type	Constraints / Description
id	UUID	PRIMARY KEY, NOT NULL
counsellor_name	VARCHAR(100)	NOT NULL
specialty	VARCHAR(100)	NOT NULL
district	VARCHAR(50)	INDEXED, NOT NULL
availability_hours	VARCHAR(100)	NOT NULL

Table 9: Columns Layout for tbl_counsellors

Section 9.10: Schema for tbl_appointments

Column Name	Data Type	Constraints / Description
id	UUID	PRIMARY KEY, NOT NULL
citizen_id	UUID	FOREIGN KEY references tbl_citizens(id)
counsellor_id	UUID	FOREIGN KEY references tbl_counsellors(id)

appointment_time	TIMESTAMP	NOT NULL
session_status	VARCHAR(30)	DEFAULT 'SCHEDULED'

Table 10: Columns Layout for tbl_appointments

Section 9.11: Schema for tbl_training_courses

Column Name	Data Type	Constraints / Description
id	UUID	PRIMARY KEY, NOT NULL
course_title	VARCHAR(100)	NOT NULL
duration_weeks	INTEGER	NOT NULL
syllabus_desc	TEXT	NOT NULL
course_fee	DECIMAL(10,2)	DEFAULT 0.00

Table 11: Columns Layout for tbl_training_courses

Section 9.12: Schema for tbl_community_posts

Column Name	Data Type	Constraints / Description
id	UUID	PRIMARY KEY, NOT NULL
author_id	UUID	FOREIGN KEY references tbl_citizens(id)
post_content	TEXT	NOT NULL
moderation_status	VARCHAR(20)	DEFAULT 'PENDING'
flag_count	INTEGER	DEFAULT 0

Table 12: Columns Layout for tbl_community_posts

Section 9.13: Schema for tbl_messages

Column Name	Data Type	Constraints / Description
id	UUID	PRIMARY KEY, NOT NULL
chat_session_id	UUID	NOT NULL
sender_type	VARCHAR(20)	e.g., BOT, CITIZEN, OFFICER
message_body	TEXT	Encrypted message text
timestamp	TIMESTAMP	DEFAULT NOW()

Table 13: Columns Layout for tbl_messages

Section 9.14: Schema for tbl_notifications

Column Name	Data Type	Constraints / Description
id	UUID	PRIMARY KEY, NOT NULL
recipient_id	UUID	FOREIGN KEY references tbl_citizens(id)
transport_channel	VARCHAR(20)	e.g., SMS, PUSH, WHATSAPP
dispatch_payload	TEXT	NOT NULL
delivery_status	VARCHAR(20)	DEFAULT 'SENT'

Table 14: Columns Layout for tbl_notifications

Section 9.15: Schema for tbl_audit_logs

Column Name	Data Type	Constraints / Description
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id	UUID	PRIMARY KEY, NOT NULL
operator_id	UUID	INDEXED, NOT NULL
transaction_type	VARCHAR(50)	NOT NULL
affected_records	VARCHAR(256)	NULL
query_payload_hash	VARCHAR(64)	SHA-256 hash
timestamp	TIMESTAMP	DEFAULT NOW()

Table 15: Columns Layout for tbl_audit_logs

Section 9.16: Schema for tbl_settings

Column Name	Data Type	Constraints / Description
param_key	VARCHAR(50)	PRIMARY KEY, NOT NULL
param_value	TEXT	NOT NULL
description	TEXT	NULL

Table 16: Columns Layout for tbl_settings

Section 9.17: Schema for tbl_media_files

Column Name	Data Type	Constraints / Description
id	UUID	PRIMARY KEY, NOT NULL
complaint_id	UUID	FOREIGN KEY references tbl_complaints(id)
file_uri	VARCHAR(512)	NOT NULL

content_type	VARCHAR(50)	NOT NULL
file_hash_sha256	VARCHAR(64)	NOT NULL

Table 17: Columns Layout for tbl_media_files

Section 9.18: Schema for tbl_activity_logs

Column Name	Data Type	Constraints / Description
id	UUID	PRIMARY KEY, NOT NULL
citizen_id	UUID	FOREIGN KEY references tbl_citizens(id)
action_tag	VARCHAR(50)	INDEXED, NOT NULL
metadata_json	TEXT	NULL

Table 18: Columns Layout for tbl_activity_logs

Chapter 10: API Reference Specifications

All interfaces expose JSON-based REST APIs. Authentication relies on OAuth2 tokens. Specific URL endpoints and parameters are to be finalized with Uttar Pradesh Police during implementation.

Chapter 11: User Stories & Acceptance Criteria

- User Story 1: As a Citizen, I want to trigger SOS silently so that police dispatchers locate me instantly.
 - Given: The citizen has configured emergency contacts.
 - When: The citizen presses the SOS button or shakes the device.
 - Then: A dispatch ticket is triggered and location telemetry is sent to nearby PCR vehicles.
- User Story 2: As a Police Station Admin (SHO), I want to review incoming voice complaints so that I can file e-FIRs.
 - Given: A voice complaint exists on the local queue.
 - When: The SHO opens the station dashboard and plays the audio.
 - Then: The translated dialect transcript is generated and verified.

Chapter 12: Security & DPDP Compliance

Data privacy is aligned with the DPDP Act 2023. Explicit consent logs are stored securely. AI capabilities are strictly assistive; all dispatch allocations, arrests, and case filings are managed by human officers. Encryption (AES-256 for PII, TLS 1.3 in transit) is enforced state-wide.

Chapter 13: UI/UX Guidelines

Interfaces follow the official police color scheme. Standard fonts (Calibri) and grid layouts ensure responsiveness across devices. High-contrast elements and speech-to-text assistants satisfy accessibility goals.

Chapter 14: Workflow Diagrams

Section 14.1: Flowchart for registration flow



Figure 1: Citizen Mobile Registration & Consent Flow

Section 14.1: Flowchart for sos flow

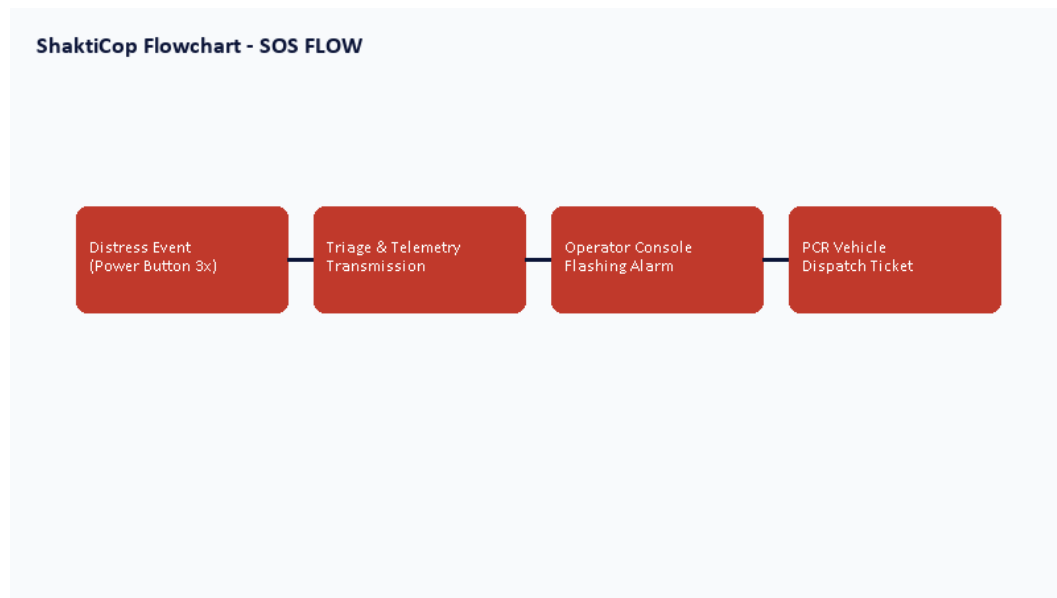


Figure 2: Silent SOS Distress & Dispatch Routing Flow

Section 14.1: Flowchart for complaint flow

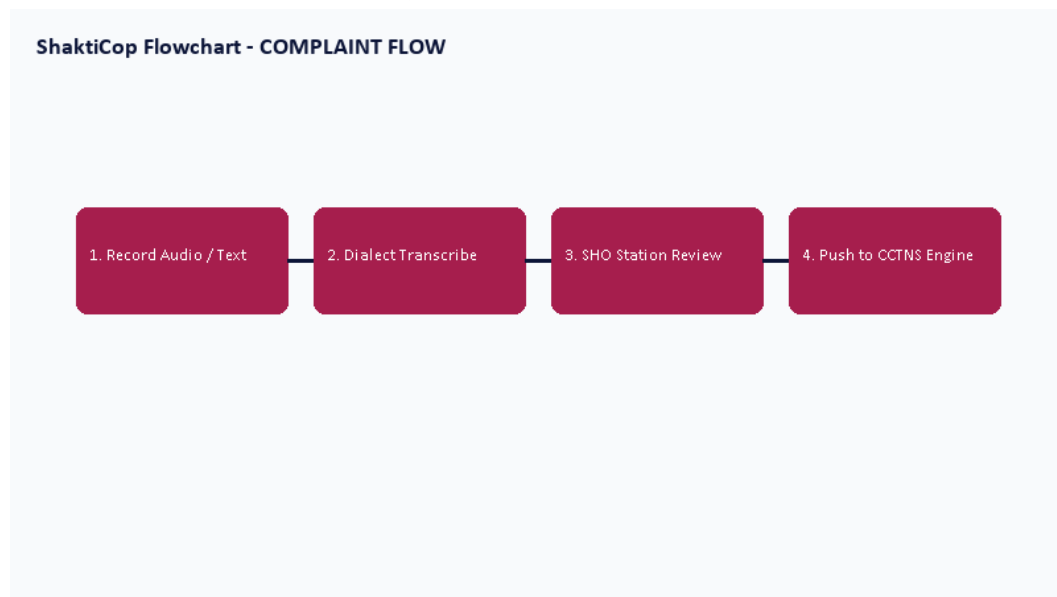


Figure 3: Voice Complaint & Dialect Transcribing Flow

Section 14.1: Flowchart for response flow

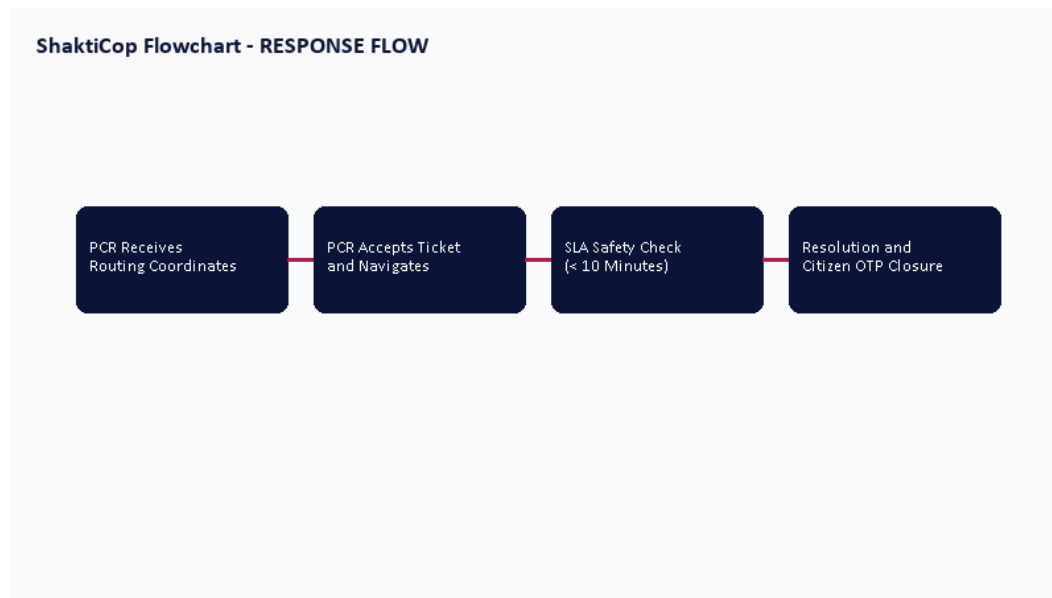


Figure 4: PCR Patrol Vehicle Dispatch & Response Flow

Chapter 15: Business Rules

1. SOS signals immediately dispatch the closest patrol unit.
2. Complaints must be reviewed by the station SHO within 24 hours.
3. Telemetry tracking automatically halts when the citizen triggers 'End Journey' and enters their OTP.

Specific threshold limits and escalation configurations are to be finalized with Uttar Pradesh Police during implementation.

Chapter 16: Technical Risks & Mitigations

Network outages, database concurrency bottlenecks, and unauthorized data access are mitigated through redundant network lines, database read replicas, and strict API firewalls.

Chapter 17: Testing & Quality Assurance

The testing strategy comprises Unit Testing, Integration Testing, System Testing, and User Acceptance Testing (UAT) conducted in coordinate segments with Uttar Pradesh Police departments.

Chapter 18: Deployment Strategy

The deployment pipeline utilizes Docker containers orchestrated via Kubernetes. Backup routines run daily. All systems are hosted at the State Data Centre (SDC) Lucknow, UP.

Chapter 19: Future Enhancements

Future phases plan the integration of Bluetooth wearable distress buttons, smart city CCTV triggers, offline mesh networks, and expanded automated vehicle routing.

Chapter 20: Conclusion

ShaktiCop establishes a modern, reliable e-Governance safety platform. Independently proposed by Adarsh Singh, this software specification lays the technical and operational foundations for review and approval by Uttar Pradesh Police.